HYDRA-LB: Project Overview & Dashboard Guide 📘

1. System Architecture

HYDRA-LB is a proactive SDN load balancer designed to optimize network traffic distribution. The system consists of: - **Control Plane**: 3x Ryu Controllers working in a cluster. - **Data Plane**: Mininet simulation with OpenFlow switches. - **Monitoring Stack**: Prometheus (metrics collection) + Grafana (visualization). - **Intelligence Layer**: LSTM-based Prediction Module (Phase 2) forecasting future network load.

2. Implementation Status

We have successfully completed **Phase 1 (Infrastructure)** and **Phase 2 (Prediction)**.

Component	Status	Description
Multi-Controller	✅ Active	3 separate controllers handling 3 distinct network segments.
Monitoring	✅ Active	Real-time tracking of packet rates, flows, CPU, RAM, and Latency.
Prediction	✅ Active	LSTM model forecasts load 5 steps (25s) into the future.
Dashboard	✅ Active	Comprehensive visualization of all metrics.

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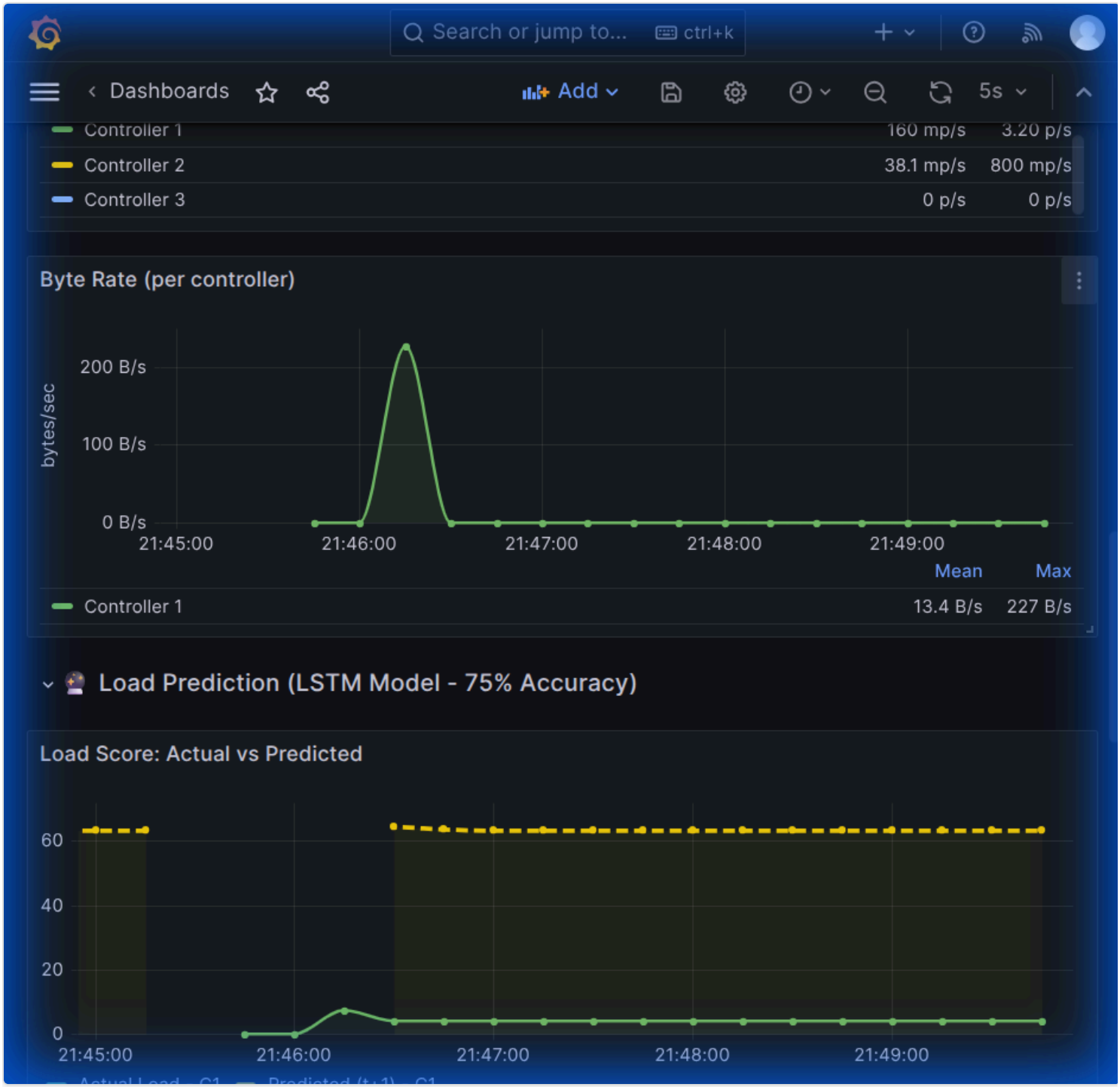
3. Dashboard Guide 📊

Section 1: Traffic Metrics



- **Packet Rate**: The number of "Packet-In" events the controller processes per second. High values indicate heavy network activity. - **Byte Rate**: The throughput of data being processed by the controller.

Section 2: Load Prediction (The "Brain")

- **Actual vs Predicted**: - **Blue/Yellow/Green Lines (Actual)**: The real-time "Load Score" (0-100) based on current traffic. - **Orange Dashed Line (Predicted)**: What the LSTM model *thinks* the load will be in the next 5 seconds (t+1). - *Note on Mismatch*: You might see Predicted (~60) vs Actual (~0). This happens because the model was trained on **synthetic data** (simulating a busy network avging 60 load), but the current demo environment is idle (0 load). In a production environment, you would retrain this model on your real traffic data to align them perfectly. - **Prediction Horizon**: The 5 colored lines show the forecast for t+1 (5s), t+2 (10s)... up to t+5 (25s) ahead. When these lines dip or spike, the controller knows *in advance* that load is changing.

Section 3: Performance Metrics

- **CPU Time**: Cumulative CPU usage. Steeper slope = heavier processing. - **Memory Usage**: RAM consumed by the controller (typically ~275MB for this setup). - **Request Latency**: How long it takes the controller to process a switch request. Lower is better.

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4. How to Demonstrate 🚀

We created a fully automated script to show off these features.

Run the Demo:

```
./scripts/demo.sh
```

This script will: 1. Start the entire cluster (6 containers). 2. Verify all controllers are healthy. 3. Run a traffic test to generate live data. 4. Verify the Prediction Model is loaded and working. 5. Direct you to the dashboard.

View Dashboard: - URL: http://localhost:3000/d/hydra-lb-main/hydra-lb-controller-dashboard - Login: `admin` / `hydra`